Name:		Centre/Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY — 9744/02

Paper 2 Structured Questions

15 September 2022

2 hours

READ THESE INSTRUCTIONS FIRST:

Write your centre number, index number, name and class at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	9
2	11
3	15
4	7
5	8
6	10
7	9
8	10
9	7
10	9
11	5
Total	100

This document consists of **24** printed pages and **2** blank pages.

Section A: Structured Questions

Answer **all** questions.

Question 1

Fig 1 shows a transmission electron micrograph (TEM) of part of a hepatocyte. Hepatocytes make up 70 to 85% of the liver's mass.

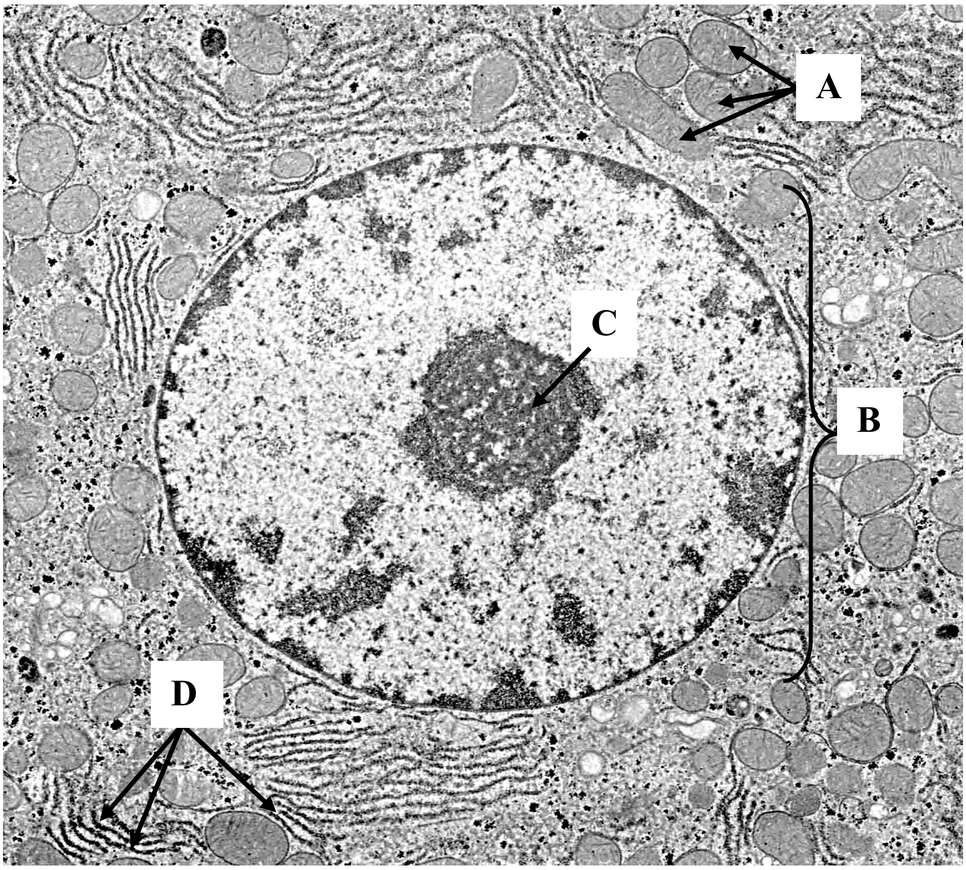

Fig 1

(a) Compare the structure of organelles A and B. [3]

(b) How are structures C and D involved in the synthesis of ribosomes? [3]

(c) Give one piece of evidence visible from Fig 1, which suggests that

(i) the cell is able to provide energy for many metabolic activities, [1]

(ii) the cell is actively synthesising proteins for use within the cell. [1]

Some bacteria secrete exotoxins out of the cell. Exotoxins are a group of soluble proteins that are capable of altering host cell physiology.

(d) Given that bacteria do not have membrane-bound organelles, suggest how the secretion of proteins in hepatocyte differs from that of bacteria. [1]

Total: [9]

Question 2

Sialidase is one of the many hydrolytic enzymes found within a lysosome. Fig 2 shows the structure of sialidase, which is a hydrolytic enzyme that is capable to digesting sialyloligosaccharides. Sialyloligosaccharides are acidic oligosaccharides that are usually made up of α-glucose with sialic acid residues linked by (2→3) and (2→6) glycosidic bonds.

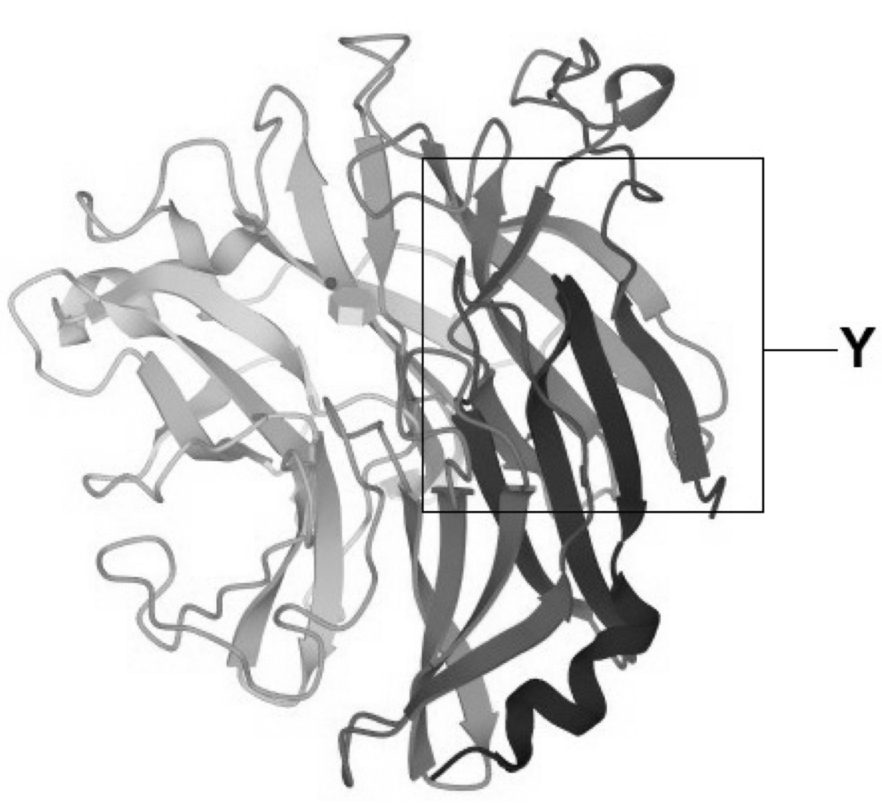

Fig 2

(a) Name structure Y and describe how it is formed. [3]

(b) Compare the structure of sialidase and its substrate. [2]

Lysosomal enzymes, such as sialidase, are only active over a narrow range of pH, with an optimum of pH 5. The pH of the cytoplasm is 7.2. The internal pH of lysosomes is maintained by concentrating H^+ ions in the lysosome.

(c) Describe how H^+ ions could be moved across the lysosomal membrane. [2]

(d) Explain why enzymes from a leaky damaged lysosome may not destroy the cell contents. [2]

(e) Suggest why the lysosomal membrane is not destroyed by the enzymes in the lysosome. [2]

Total: [11]

Question 3

Fig 3.1 shows cells of an organism in various stages of mitosis.

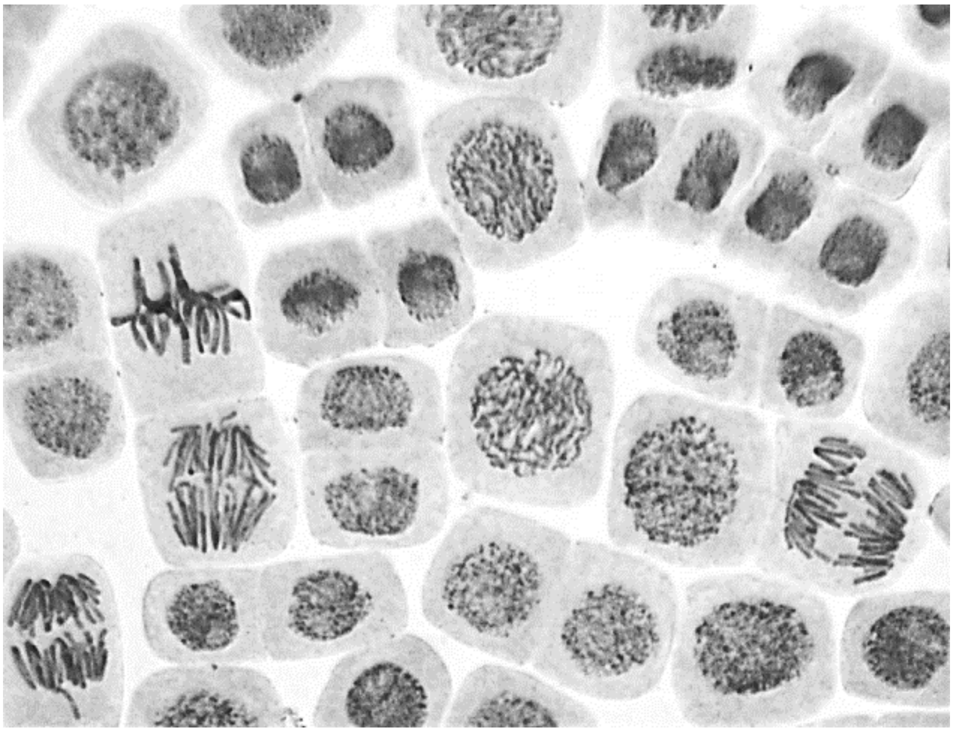

Fig 3.1

(a) On Fig 3.1, use labels and label lines to indicate one cell in prophase and anaphase. [1]

(b) Describe how the mitotic cell cycle is controlled to ensure the production of daughter cells that are genetically identical to parental cells. [3]

Cancer occurs when the cells escape the control mechanism that normally limits their proliferation, and thus leads to uncontrolled proliferation of cells. This results in tumorigenesis.

There are different types of chemotherapeutic drugs to treat cancer. One example is cisplatin, which is used for the treatment of testicular and ovarian cancer. Each cisplatin molecule forms two chemical bonds with a DNA molecule. Fig 3.2 shows two ways in which cisplatin can bind to the DNA molecule.

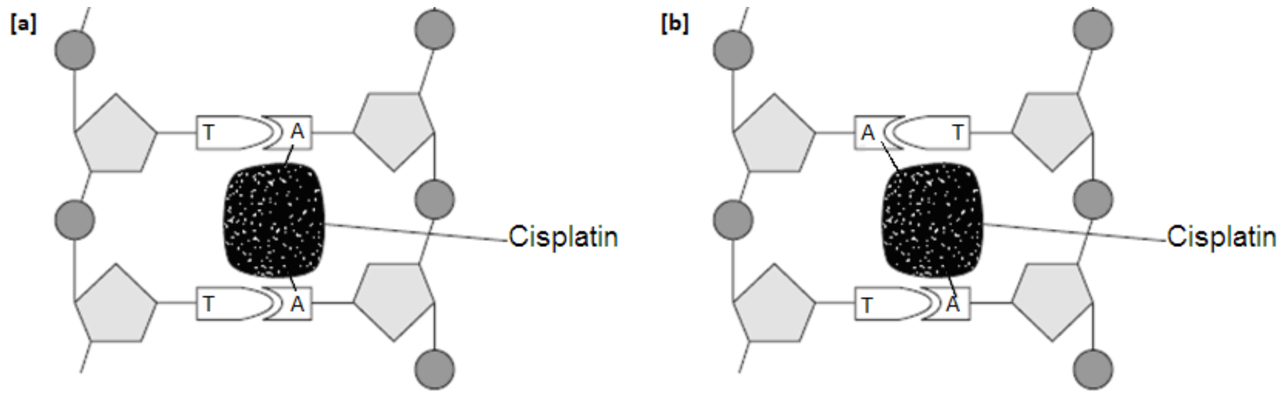

Fig 3.2

(c) **(i)** State precisely the cell cycle phase that cisplatin acts on. [1]

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(ii) Explain how cisplatin inhibits the division of cancer cells. [3]

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While cisplatin is one of the most powerful drugs for testicular and ovarian cancer, it can interfere with the function of normal cells and cause severe side effects. Hence, scientists have been creating alternative drugs in attempts to find treatments without side effects. One approach is to develop light-activated drugs.

With a team of researchers, Thorn-Seshold developed a method for optically controlling microtubule inhibitor drugs. The strategy involves identifying a fixed structural element, which is required for a drug's biological activity, and then replacing that element with a flexible hinge that swings open or shuts in response to blue light. Hence, blue light can be used to switch the hinged drug on and off with single-cell precision.

(d) **(i)** State two functions of microtubule in a normal, non-dividing cell. [2]

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(ii) Explain how interfering with the function of microtubules helps to treat cancer. [3]

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(iii) Suggest how the change in the flexible hinge of the drug can be optically controlled. [2]

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Total: [15]

Question 4

Fig 4 shows a DNA replication fork.

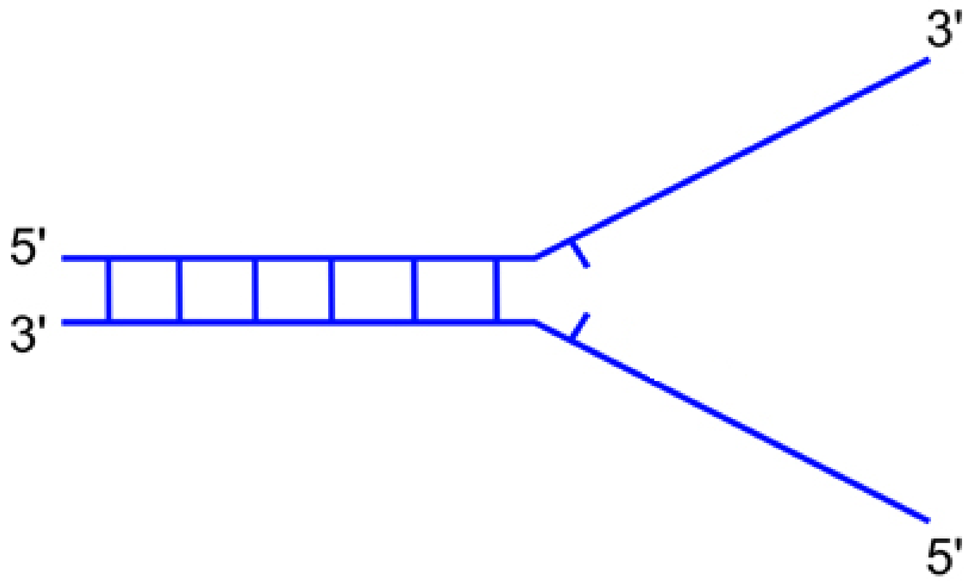

Fig 4

(a) (i) Using two arrows (→), draw the newly-synthesized Okazaki fragments on Fig 4. [1]

(ii) Explain why synthesis of DNA occurs both continuously and discontinuously during DNA replication. [2]

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(b) Outline the steps that occur to produce a complete DNA molecule after the formation of Okazaki fragments. [3]

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(c) Describe the importance of complementary base pairing in DNA replication. [1]

Total: [7]

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Question 5

In the 1950s, Erwin Chargaff determined the relative quantities of the four bases in DNA in different organisms. His results provided important evidence for the model of DNA proposed by James Watson and Francis Crick in 1953. Table 5.1 shows the relative quantities of the four bases in bacteria, yeast and an unknown.

Table 5.1

Sample	Percentage of adenine	Percentage of thymine	Percentage of guanine	Percentage of cytosine
Escherichia coli	24.7	23.6	26.0	25.7
yeast	31.3	32.9	18.7	17.1
unknown	24.0	31.2	23.3	21.5

(a) Explain why the result of the unknown differs from those of yeast and bacteria? [2]

Protein synthesis in both eukaryotic and prokaryotic cell is a multi-step process.

(b) Suggest how protein synthesis in eukaryotes will be affected when there is a mutation to the TATA region on DNA. [2]

Fig 5 shows the base sequence on a segment of mRNA coding for Ras protein, with their corresponding amino acids.

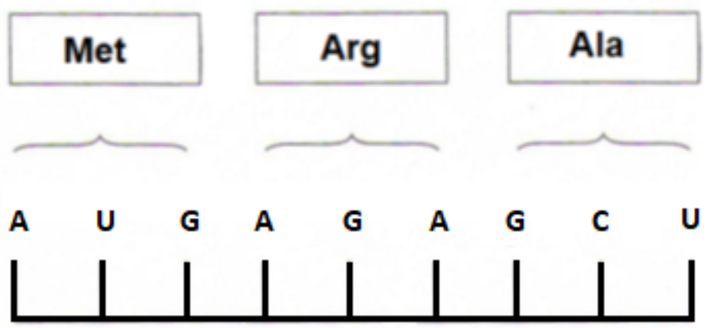

Fig 5

mRNA sequences from ten closely related species of rats were analysed. It was found that some regions are highly conserved. However, the third base position of many codons is least conserved. For example, amino acid arginine can be coded by AGA and AGG codons.

(c) **(i)** Explain why the third base position of this codon is less conserved than the first and second positions. [2]

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(ii) The codons in the mRNA sequence do not overlap. Suggest an advantage of this. [1]

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Some genetic drugs are short sequences of nucleotides. They act by binding to selected sites on DNA or mRNA molecules and preventing the synthesis of disease-related proteins. The two types of drugs include:

1. Antisense drugs made from ribonucleotides that bind to mRNA.
2. Triplex drugs made from deoxyribonucleotides that bind to DNA forming a triple-helix.

Table 5.2 below shows the sequence of bases on the part of a molecule on mRNA.

Table 5.2

Sequence of part of mRNA	Sequence of part of antisense drug
A	
C	
G	
U	
G	

(d) Complete Table 5.2 to show the base sequence on the antisense drug that binds to this mRNA. [1]

Total: [8]

Question 6

Fig 6 shows a type of virus called a T2 phage. It consists of a protein coat enclosing a DNA molecule. This DNA causes the bacterial cell to make new copies of the T2 phage.

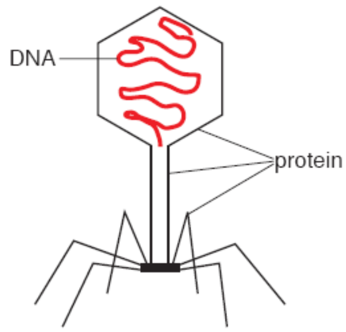

Fig 6

(a) **(i)** State two ways in which the structure of the protein coat of T2 differs from the structure of its DNA. [2]

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(ii) List three features of a bacterium that T2 does not share. [3]

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(iii) Describe what happen after T2 phage genome enters bacterium host to complete its life cycle. [3]

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(b) T2 phages were used in experiments in 1952 to find out whether protein or DNA is the genetic material responsible for inheritance. The protein molecules in the coat were labelled with a radioactive isotope of sulphur, ^{35}S and the DNA inside was labelled with a radioactive isotope of phosphorus, ^{32}P.

(i) Name the bond in protein that joins two sulphur atoms. [1]

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(ii) Name the part of the deoxyribonucleotide to which phosphorus atoms are bonded. [1]

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Total: [10]

Question 7

Fig 7 shows a segment of the *Escherichia coli* chromosome that contains the *lac* operon.

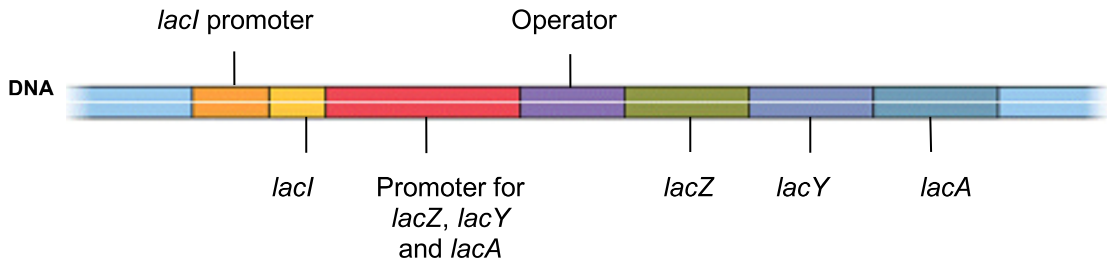

Fig 7

(a) With reference to Fig 7,

(i) identify one coding and one non-coding DNA sequence, [1]

Coding DNA: ....

Non-coding DNA: ....

(ii) describe three differences between the organisation of eukaryotic and prokaryotic genes. [3]

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The gene activity of the *lac* operon in four different *E. coli* strains was measured and the results are shown in Table 7.

Table 7

***E. coli* strain**	**Genotype**	**Presence (✓) or absence (×) of β-galactosidase activity**	
		Lactose present	**Lactose absent**
E1	$I^+ O^+ Z^+$	✓	×
E2	$I^- O^+ Z^+$	✓	✓
E3	$I^+ O^c Z^+$	✓	✓
E4	$I^S O^+ Z^+$	×	×

Key	
+	wild type
-, c, s	mutant

(b) Using information from Table 7, account for the activity of β-galactosidase in strains E2 and E3 in terms of the expression of LacI protein and its binding to the operator region. [2]

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(c) The activity of β-galactosidase in E4 was due to a single-base substitution mutation in the lacI gene, resulting in a change in the LacI protein. [3]

Explain how this mutation could have resulted in the change.

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Total: [9]

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Question 8

In *Drosophilia* flies, the gene for wing shape is found on chromosome 2. Allele C for curly wing is dominant to allele c for normal wing.

Female flies are homogametic while males are heterogametic. The gene for eye colour is found on the X chromosome. Allele for red eyes is dominant to the allele for white eyes.

(a) **(i)** Fill in the blanks using suitable symbols. [1]

	Female fly	**Male fly**
Eye colour	White	Red
Wing shape	Curly wing, heterozygous	Curly wing, heterozygous
Genotype		

(ii) Draw a genetic diagram showing the phenotypes of the offspring and the expected phenotypic ratio when these two flies are crossed. [4]

(b) 240 offspring were obtained from this cross and the table below shows the number of offspring for each phenotype.

Red-eyes, curly wings female	83
Red-eyes, normal wings female	41
White-eyes, curly wings male	78
White-eyes, normal wings male	38

	Probability				
Degrees of freedom	**0.90**	**0.50**	**0.10**	**0.05**	**0.01**
2	0.21	1.39	4.61	5.99	9.21
3	0.58	2.37	6.25	7.81	11.34
4	1.06	3.36	7.78	9.49	13.28
5	1.61	4.35	9.24	11.07	15.09
6	2.20	5.35	10.64	12.59	16.81

Formula for χ^2 calculation

$$\chi^2 = \Sigma \frac{(O-E)^2}{E} \qquad v = c - 1$$

where Σ = 'sum of…' O = observed 'value'
v = degrees of freedom E = expected 'value'
c = number of classes

Complete the table to obtain the calculated χ^2 value and state what conclusion may be drawn from the result. Show all the necessary workings and calculations. [5]

Classes	***Observed number (O)***	***Expected ratio***	***Expected number (E)***	**$(O - E)^2$**	**$\frac{(0 - E)^2}{E}$**
Red eyes, curly wings female					
Red-eyes, normal wings female					
White-eyes, curly wings male					
White-eyes, normal wings male					
					$\sum\chi^2 =$ ________

Total: [10]

Question 9

2,4-dinitrophenol is a chemical that is toxic to mitochondria. When added to mitochondria, this chemical allows electron transport to occur but prevents the phosphorylation of ADP to ATP. The chemical achieves this by breaking the essential link between electron transport and ATP synthesis. This toxin causes mitochondria to produce heat instead of ATP. The greater the amount of toxin added, the faster is its action.

(a) If mitochondria are poisoned with 2,4-dinitrophenol, by what process could a plant cell produce more ATP? [1]

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(b) A researcher wanted to study cellular respiration in insect cells. She cultured some muscle cells from the common field cricket, *Teleogryllus oceanicus*, and then studied the effects of adding 2,4-dinitrophenol to these cells. An agricultural company was interested to fund this research. [1]

Suggest why an agricultural company might want to fund research on the effects of this toxin on field crickets.

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The experiment is summarized in Table 9. Temperature observations in each trial were made at equal time intervals.

Table 9

Observations made at equal time intervals	**Temperature / °C**		
	Control (no 2,4-dinitrophenol)	**Trial 1 (2,4-dinitrophenol added)**	**Trial 2**
1 (start)	28	28	28
2	27	28	29
3	28	29	
4	29	31	
5	28	36	
6	28	23	
7	27	21	

(c) Explain why the temperature went down after the fifth observation in trial 1. [1]

(d) Trial 2 had twice the concentration of 2,4-dinitrophenol added. [2]

Complete Table 9 by writing temperatures in the spaces provided to predict the trend.

(e) Another researcher suggested adding pyruvate to the cells at the beginning of the experiment to cancel out the effects of this toxin. [2]

Explain what effect adding pyruvate would have on cancelling out the effects of this toxin.

Total: [7]

Question 10

Fig 10 shows the action of the hormone insulin on liver cells to regulate the concentration of blood glucose.

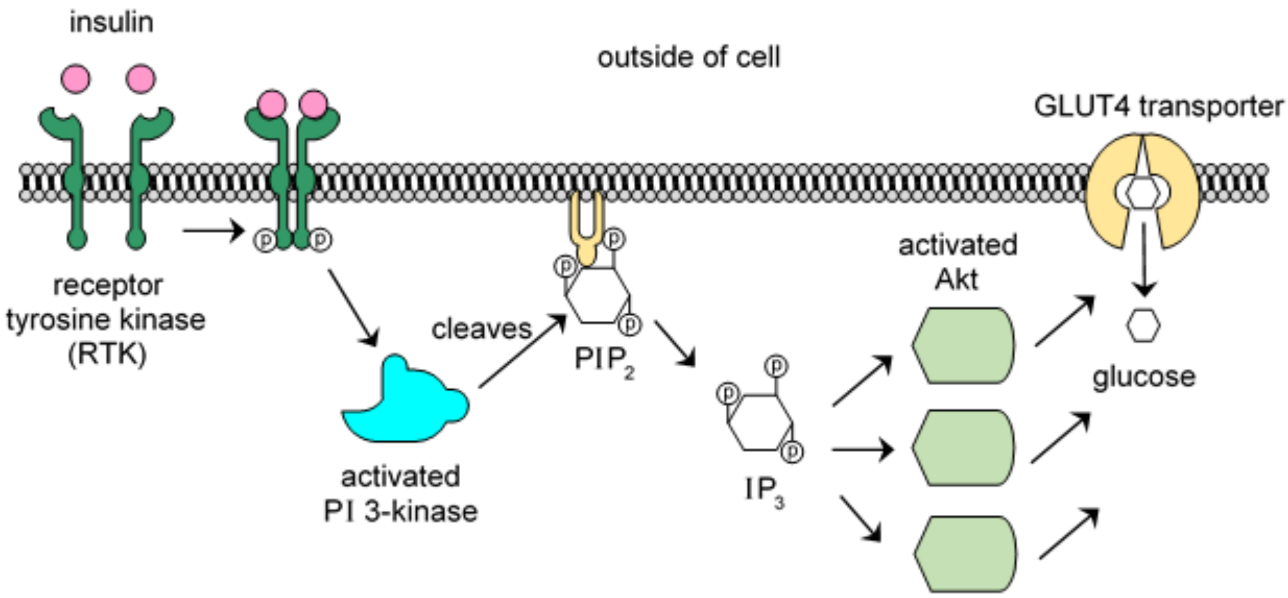

Fig 10

(a) Outline the process of insulin-receptor interaction. [4]

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(b) Describe the nature of IP_3 and explain its significance in insulin signalling. [3]

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(c) Explain how GLUT4 transporters regulate the concentration of blood glucose. [2]

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Total: [9]

Question 11

Table 11 shows the range of optimal temperatures for the growth of some major crops in the state of Iowa, USA. These crops are generally grown in April and harvested in October where average temperatures are between 16ºC and 28ºC.

Table 11

Type of crop	Temperature/ ºC		
	Optimal	Minimum	Maximum
Corn	22 - 25	20	32 - 34
Wheat	20 - 25	5	38
Soybean	25 - 28	10 -14	37 - 40

(a) With reference to Table 11 and your knowledge of enzymes, explain why Iowa is suitable for the growth of these crops. [2]

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(b) Suggest why corn productivity may be threatened by climate change. [3]

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Total: [5]